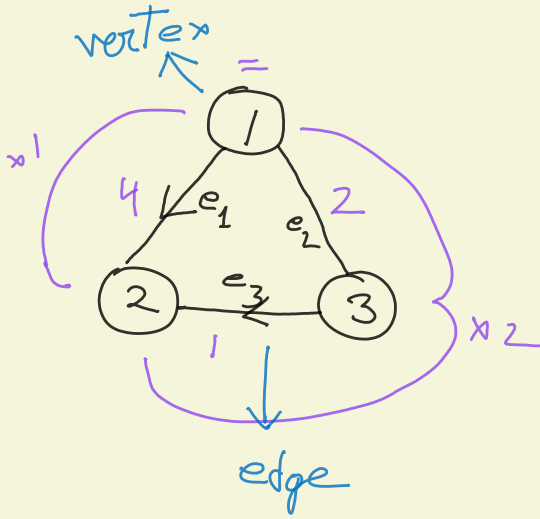


Graph :- A set representation in order is a graph.

$$A = \{ 3, 6, 9, 10, 11, \dots, n \} \rightarrow \text{set}$$



- ① with direction ✓
- ② without direction ✓
- ③ with weight ✓

$$G = \{ (1, 2)^{e_1}, (1, 3)^{e_2}, (3, 2)^{e_3}, (3, 1)^{e_4} \}$$

Graph Representation

↳ Matrix $\rightarrow$ loss of memory $a[i][j]$

↳ List (Adjacency List) $\rightarrow$ memory efficient

Matrix Representation

$$G = \{ (1, 2), (2, 3), (1, 4) \}$$

	1	2	3	4
1	0	1	0	3
2	1	0	4	0
3	0	1	0	0
4	1	0	0	0

0 $\rightarrow$ no edge
1 $\rightarrow$ edge present

$$a[1][2] = 1$$

$$a[2][3] = 5$$

$$a[7][8] = 0$$

Concepts-

- Dense Graph
- Sparse Graph